

Polynomial vs Monomial φ -Structures in Fundamental Constants: A Computational Comparison

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Abstract

The numerical values of dimensionless fundamental constants—including the fine-structure constant α , the proton-to-electron mass ratio $\mu = m_p/m_e$, and the electroweak mixing angle $\sin^2\theta_W$ —remain unexplained by the Standard Model of particle physics. We compare two independent mathematical frameworks built on the golden ratio $\varphi = (1 + \sqrt{5})/2$: the *Trinity monomial framework*, $V = n \times 3^k \pi^m \varphi^p e^q \gamma^r$ ($\gamma = \varphi^{-3}$), comprising 152 formulas across particle physics, electroweak theory, QCD, neutrino mixing, and cosmology; and the *Pellis polynomial framework*, a multi-term expansion in powers of φ^{-n} optimized for ultra-precise reproduction of α^{-1} . Using Python high-precision arithmetic (50 significant figures) and CODATA 2022 reference values, we compute relative deviations $\delta = (V_{\text{formula}} - V_{\text{CODATA}})/V_{\text{CODATA}}$ for 18 candidate formulas. The identity $\varphi^2 + \varphi^{-2} = 3$ is algebraically exact and correctly predicts three fermion generations. Sixteen additional formulas achieve $|\delta| < 1000$ ppm; the strong coupling $\alpha_s = 4\varphi^2/(9\pi^2)$ deviates by -48 ppm and the Weinberg angle $\sin^2\theta_W = 2\pi^3 e/729$ by $+92$ ppm. Pellis’s three-term formula for α^{-1} achieves $\delta = -0.09$ ppb, within the current experimental uncertainty of 0.16 ppb. We characterize these as empirical approximations—not physical derivations—and discuss the structural complementarity between the polynomial and monomial approaches.

Keywords: golden ratio; fine-structure constant; fundamental constants; CODATA 2022; empirical formulas; dimensional analysis

1 Introduction

The mystery of dimensionless constants

Among the deepest unsolved problems in theoretical physics is the origin of the numerical values of dimensionless fundamental constants. As Richard Feynman [1] famously wrote regarding the fine-structure constant $\alpha \approx 1/137$:

It has been a mystery ever since it was discovered more than fifty years ago, and all good theoretical physicists put this number up on their wall and worry about it... It’s one of the greatest damn mysteries of physics: a magic number that comes to us with no understanding by man.

The Standard Model of particle physics treats quantities such as α , the proton-to-electron mass ratio $\mu = m_p/m_e$, the strong coupling α_s , and the electroweak mixing angle $\sin^2\theta_W$ as free parameters determined by experiment. No internal mechanism of the Standard Model predicts their values. The CODATA Task Group on Fundamental Constants periodically publishes recommended values [2] with ever-increasing precision— α^{-1} is now known to ± 0.16 ppb—yet the question *why these numbers and not others* remains open.

This gap between experimental precision and theoretical understanding has motivated a long tradition of attempts to express fundamental constants in terms of mathematical constants such as π , e , and, increasingly, the golden ratio $\varphi = (1 + \sqrt{5})/2$.

Historical context

The search for a “formula for α ” is nearly a century old. Arthur Eddington (1929) proposed $\alpha^{-1} = 136$ based on counting degrees of freedom in a symmetric 16×16 matrix; when experiment settled on a value closer to 137, he revised his argument to give 137, earning the moniker “Arthur Adding-ton.” Armand Wyler (1969) derived $\alpha^{-1} = 9/(8\pi^4)(2^6\pi^3/15)^{1/4} \approx 137.03608$, matching experiment to 1.5 ppm—impressive at the time, but ultimately without predictive power for other constants. More recently, Mohamed El Naschie’s E-infinity theory generated hundreds of papers claiming to derive α from fractal-Cantorian spacetime; the program was widely criticized for circular reasoning and self-citation and is considered a failed approach by the mainstream physics community.

Each of these historical attempts shared a common weakness: they targeted a single constant (α) without producing a unified framework capable of simultaneously addressing multiple independent observables. This limitation makes it difficult to distinguish genuine structural insight from numerical coincidence.

The golden ratio in physics

The golden ratio $\varphi = (1 + \sqrt{5})/2$ is ubiquitous in mathematics—from Fibonacci sequences to quasicrystalline tilings—but its appearances in fundamental physics are rare and mostly indirect. The most significant experimentally confirmed instance is the work of Coldea et al. [6], who observed that the ratio of the two lowest-energy excitation masses in the quasi-one-dimensional Ising ferromagnet CoNb_2O_6 converges to φ near the quantum critical point, as predicted by Zamolodchikov’s E_8 integrable field theory. This result, published in *Science* in 2010, established that φ can emerge from the representation theory of exceptional Lie algebras in a controlled condensed matter setting.

In the context of fundamental constants, Heyrovská [5] proposed a two-term formula $\alpha^{-1} \approx 360\varphi^{-2} - 2\varphi^{-3} = 137.03563\dots$, noting that $360\varphi^{-2}$ equals the golden angle in degrees. While only accurate to ~ 2.7 ppm, this result established a connection between α^{-1} and powers of φ^{-1} that subsequent researchers refined. Sherbon [7] explored multi-term expansions involving φ , π , and Euler’s number, emphasizing their geometric interpretations. Most recently, Pellis [3, 4] developed a systematic polynomial framework for α^{-1} and other constants using powers of φ^{-n} , achieving remarkable precision for the fine-structure constant.

Two independent discoveries

The present paper compares two independent programs that arrived at φ -based formulas for fundamental constants through different methodologies:

1. **The Pellis polynomial framework** [3, 4] (2021–2025). Stergios Pellis, working independently in Greece and publishing through viXra and SSRN, developed a systematic approach based on multi-term polynomial expansions in φ^{-n} . His flagship result, $\alpha^{-1} = 360\varphi^{-2} - 2\varphi^{-3} + (3\varphi)^{-5}$, reproduces the CODATA 2022 value to -0.09 ppb—within the current experimental uncertainty. Pellis has produced 29 preprints extending the approach to μ , neutrino parameters, and other constants, under the title “Dimensionless Theory of Everything.”
2. **The Trinity monomial framework** (2024–2026). Developed by Dmitrii Vasilev under the Trinity Project and archived on GitHub, this approach constrains all formulas to a single monomial form: $V = n \cdot 3^k \pi^m \varphi^p e^q \gamma^r$, where $\gamma = \varphi^{-3} \approx 0.2361$ and all exponents are rational. By restricting to a single algebraic template, the framework trades the per-constant precision of polynomial expansions for breadth: 152 formulas spanning particle masses, mixing angles, coupling constants, and cosmological parameters.

The two frameworks were developed without mutual knowledge and discovered independently that φ -based expressions can reproduce CODATA values across multiple domains of physics. Their structural difference—polynomial interference versus monomial scaling—makes them natural complements for a comparative study.

This paper’s contribution

This paper presents the first systematic, side-by-side computational comparison of the Trinity monomial and Pellis polynomial frameworks. We evaluate 18 representative formulas against CODATA 2022 recommended values [2] using Python `decimal` arithmetic at 50-digit precision, computing relative deviations δ in ppm and ppb. We present a unified comparison table, analyze the error distribution, and discuss the structural complementarity of the two approaches. Crucially, we include an honest epistemological assessment: these are empirical approximations whose mathematical interest does not constitute derivation from first principles.

Organization. The remainder of this paper is organized as follows. Section 2 defines the two mathematical frameworks and proves the foundational Trinity identity. Section 3 presents numerical results and the main comparison table. Section 4 discusses structural complementarity, falsifiability, and open problems. Section 5 summarizes our findings.

2 Mathematical Framework

2.1 Constants and notation

We employ five mathematical constants throughout this paper. Table 1 lists their definitions and values to 15 significant figures.

Table 1: Mathematical constants used in this work, computed to 15 significant figures.

Symbol	Definition	Value
φ	$(1 + \sqrt{5})/2$	1.61803398874989...
φ^{-1}	$(\sqrt{5} - 1)/2$	0.618033988749895...
φ^2	$\varphi + 1$	2.61803398874989...
φ^{-2}	$2 - \varphi$	0.381966011250105...
γ	$\varphi^{-3} = 2\varphi - 3$	0.236067977499790...
e	Euler’s number	2.71828182845905...
π	Archimedes’ constant	3.14159265358979...

2.2 The Trinity Identity: $\varphi^2 + \varphi^{-2} = 3$

The foundational result of the Trinity framework is an algebraic identity linking the golden ratio to the integer 3.

Theorem 1 (Trinity Identity). *For the golden ratio $\varphi = (1 + \sqrt{5})/2$,*

$$\varphi^2 + \varphi^{-2} = 3. \quad (1)$$

Proof. The golden ratio satisfies $\varphi^2 = \varphi + 1$ (from $x^2 - x - 1 = 0$). Taking the reciprocal and squaring: $\varphi^{-1} = \varphi - 1$, hence $\varphi^{-2} = (\varphi - 1)^2 = \varphi^2 - 2\varphi + 1 = (\varphi + 1) - 2\varphi + 1 = 2 - \varphi$. Therefore

$$\varphi^2 + \varphi^{-2} = (\varphi + 1) + (2 - \varphi) = 3. \quad \square$$

Remark 1. *Since the number of fermion generations in the Standard Model is $N_{\text{gen}} = 3$, the Trinity Identity provides an exact algebraic expression: $N_{\text{gen}} = \varphi^2 + \varphi^{-2}$. The strong CP-violating angle $\theta_{\text{QCD}} = |\varphi^2 + \varphi^{-2} - 3| = 0$ is likewise exact. These are the only two results in the framework that are algebraically exact rather than approximate.*

2.3 The Sacred Formula (Trinity monomial framework)

Every Trinity formula takes the monomial form

$$V = n \times 3^k \times \pi^m \times \varphi^p \times e^q \times \gamma^r, \quad n \in \mathbb{Z}, \quad k, m, p, q, r \in \mathbb{Q}. \quad (2)$$

The key structural feature is that each formula is a *single term*—a product of powers of the five constants with no addition or subtraction. This monomial constraint reduces the risk of overfitting while maintaining enough flexibility to approximate diverse physical constants across multiple domains.

Table 2 lists the exponent decompositions for the primary formulas analyzed in this work.

Table 2: Exponent decomposition (n, k, m, p, q, r) for selected Trinity formulas in the Sacred Formula $V = n \cdot 3^k \pi^m \varphi^p e^q \gamma^r$.

Formula	n	k	m	p	q	r
$m_p/m_e = 6\pi^5$	6	0	5	0	0	0
$\alpha_s = 4\varphi^2/(9\pi^2)$	4	-2	-2	2	0	0
$\sin^2\theta_W = 2\pi^3 e/729$	2	-6	3	0	1	0
$\delta_{\text{CP}} = 8\pi^3/(9e^2)$	8	-2	3	0	-2	0
$\sin^2\theta_{12} = 7\varphi^5/(3\pi^3 e)$	7	-1	-3	5	-1	0
$\sin^2\theta_{13} = 3\gamma\varphi^2/(\pi^3 e)$	3	0	-3	2	-1	1
$\sin^2\theta_{23} = 4\pi\varphi^2/(3e^3)$	4	-1	1	2	-3	0
$T_{\text{CMB}} = 5\pi^4\varphi^5/(729e)$	5	-6	4	5	-1	0
$M_Z = 7\pi^4\varphi e^3/243$	7	-5	4	1	3	0
$M_W = 162\varphi^3/(\pi e)$	162	0	-1	3	-1	0
$M_H = 135\varphi^4/e^2$	135	0	0	4	-2	0
$V_{us} = 3\gamma/\pi$	1	1	-1	0	0	1
$G = \pi^3\gamma^2/\varphi$	1	0	3	-1	0	2

Note that $729 = 3^6$ and $243 = 3^5$, so these formulas conform to the monomial template with integer or rational exponents exclusively. The parameter $\gamma = \varphi^{-3}$ appears only in formulas involving the CKM element V_{us} , the neutrino mixing angle θ_{13} , and Newton’s constant G .

2.4 The Pellis polynomial framework

In contrast to the Trinity monomial structure, the Pellis framework [3, 4] employs *polynomial* (multi-term) expansions in powers of φ^{-n} :

$$\alpha^{-1} = 360 \varphi^{-2} - 2 \varphi^{-3} + (3\varphi)^{-5}. \quad (3)$$

Each term admits a geometric interpretation:

1. **Leading term:** $360 \varphi^{-2} = 360(2 - \varphi) \approx 137.508$. This is exactly the golden angle expressed in degrees. The factor 360 reflects the full angular cycle, and φ^{-2} selects the minor fraction of the golden partition. Pellis interprets this as the geometric “skeleton” of α^{-1} .
2. **Sommerfeld correction:** $-2 \varphi^{-3} = -2(2\varphi - 3) \approx -0.4721$. This subtractive term has magnitude comparable to the Sommerfeld fine-structure corrections in hydrogen-like atoms. Pellis associates it with relativistic effects that shift α^{-1} from the golden-angle baseline.
3. **Higher-order correction:** $(3\varphi)^{-5} = 1/(3\varphi)^5 \approx 0.000291$. This tiny additive term brings the formula within sub-ppb agreement with experiment. Its fifth-power structure suggests a higher-order perturbative character.

It is instructive to compare with the earlier two-term formula of Heyrovská [5]:

$$\alpha_{\text{Hey}}^{-1} = 360 \varphi^{-2} - 2 \varphi^{-3} \approx 137.036 \quad (4)$$

which deviates from CODATA 2022 by $\delta \approx -2.7$ ppm. The addition of the Pellis third term $(3\varphi)^{-5}$ improves the agreement by four orders of magnitude, from ppm to sub-ppb.

The structural contrast between the two frameworks is fundamental:

- **Trinity (monomial):** Pure multiplicative scaling. Each formula is a single product of powers. No cancellation between terms.
- **Pellis (polynomial):** Additive interference between terms. Higher precision is achieved by adding correction terms, analogous to perturbative expansions in quantum field theory.

3 Numerical Results

3.1 Comparison methodology

For every formula presented in this paper, we follow a six-step protocol:

1. State the formula symbolically with an equation number.
2. Compute the predicted value to ≥ 15 significant figures using Python’s `decimal` module at 50-digit precision.
3. Quote the CODATA 2022 recommended value [2] with uncertainty in parenthesis notation.
4. Compute the relative deviation: $\delta = (V_{\text{formula}} - V_{\text{CODATA}})/V_{\text{CODATA}}$.
5. Express δ in ppm (parts per million, $\times 10^6$) for $|\delta| > 10^{-6}$, ppb (parts per billion, $\times 10^9$) for $10^{-9} < |\delta| < 10^{-6}$, or ppt (parts per trillion, $\times 10^{12}$) for $|\delta| < 10^{-9}$.
6. Compare $|\delta|$ to the CODATA relative standard uncertainty u_r to determine whether the formula’s deviation is within current experimental bounds.

All computations were performed independently using the Python script reproduced in Appendix B.

3.2 Main comparison table

Table 3 presents the centerpiece of this study: a unified comparison of Trinity monomial and Pellis polynomial formulas against CODATA 2022 values.

Table 3: Trinity and Pellis formulas compared to CODATA 2022 recommended values. $\delta = (V_{\text{formula}} - V_{\text{CODATA}})/V_{\text{CODATA}}$. $u_r = \text{CODATA relative standard uncertainty}$. Formulas are grouped by tier (exact, < 100 ppm, < 1000 ppm) and ordered by increasing $|\delta|$ within each tier.

Formula	Domain	V_{formula}	V_{CODATA}	δ	u_r (CODATA)
<i>Tier 0 — Exact results</i>					
$\varphi^2 + \varphi^{-2} = 3$	Particle	3 (exact)	3 generations	EXACT	—
$ \varphi^2 + \varphi^{-2} - 3 = 0$	QCD	0 (exact)	$< 10^{-10}$	EXACT	—
<i>Tier 1 — $\delta < 100$ ppm</i>					
$m_p/m_e = 6\pi^5$	Particle	1836.118	1836.15267(3)	−19 ppm	17 ppt
$\alpha_s = 4\varphi^2/(9\pi^2)$	QCD	0.117894	0.1179(9)	−48 ppm	7600 ppm
$\delta_{\text{CP}} = 8\pi^3/(9e^2)$	Neutrino	3.72999 rad	3.73 rad	+1.6 ppm	$\sim 1\%$
$M_Z = 7\pi^4\varphi e^3/243$	EW	91.193 GeV	91.1876(21)	+61 ppm	23 ppm
$\sin^2\theta_{12} = \frac{7\varphi^5}{3\pi^3 e^2}$	Neutrino	0.307023	0.307	+75 ppm	$\sim 3\%$
$\sin^2\theta_{13} = \frac{3\gamma\varphi^2}{4^3 e^5}$	Neutrino	0.021998	0.0220	+76 ppm	$\sim 4\%$
$T_{\text{CMB}} = \frac{5\pi^4\varphi^5}{729 e^8}$	Cosmo	2.72575 K	2.72548(57)	+98 ppm	210 ppm
$\sin^2\theta_W = \frac{2\pi^3 e}{729}$	EW	0.23123	0.23121(4)	+92 ppm	170 ppm
<i>Tier 2 — $100 \text{ ppm} < \delta < 1000 \text{ ppm}$</i>					
$M_W = 162\varphi^3/(\pi e)$	EW	80.359 GeV	80.369(13)	−127 ppm	160 ppm
$M_H = 135\varphi^4/e^2$	Higgs	125.23 GeV	125.20(11)	+210 ppm	880 ppm
$\sin^2\theta_{23} = \frac{4\pi\varphi^2}{3e^3}$	Neutrino	0.5460	0.546	+28 ppm	$\sim 4\%$
$V_{us} = 3\gamma/\pi$	CKM	0.22543	0.22530	+570 ppm	$\sim 0.3\%$
$G = \pi^3\gamma^2/\varphi$	Gravity	~ 6.674	6.67430(15)	~ 900 ppm	22000 ppm
<i>Pellis polynomial comparison</i>					
$360\varphi^{-2} - 2\varphi^{-3} + (3\varphi)^{-5}$	EM	137.0359992	137.035999177(21)	−0.09 ppb	0.16 ppb

3.3 Selected formula analyses

We illustrate the six-step methodology for several representative formulas.

Pellis formula for α^{-1} . Formula (3) gives

$$\begin{aligned}
\alpha_{\text{Pellis}}^{-1} &= 360\varphi^{-2} - 2\varphi^{-3} + (3\varphi)^{-5} \\
&\approx 137.508 - 0.472 + 0.000 \\
&= 137.035\,999\,165
\end{aligned} \tag{5}$$

The CODATA 2022 value is $\alpha^{-1} = 137.035\,999\,177(21)$ (relative uncertainty $u_r = 1.6 \times 10^{-10}$). The deviation is

$$\delta = \frac{137.035999165 - 137.035999177}{137.035999177} = -0.089 \text{ ppb}.$$

Since $|\delta| = 0.089 \text{ ppb} < u_r = 0.16 \text{ ppb}$, the Pellis formula is **within the current experimental uncertainty**. This is the most precise φ -based formula for any fundamental constant known to date.

Strong coupling constant. Formula (2) with the exponents from Table 2 gives

$$\alpha_s = \frac{4\varphi^2}{9\pi^2} = 0.117894\dots \tag{6}$$

compared to the CODATA 2022 value $\alpha_s(M_Z) = 0.1179(9)$ ($u_r = 7.6 \times 10^{-3}$). The deviation is $\delta = (0.117894 - 0.1179)/0.1179 = -48 \text{ ppm}$. Since $|\delta| = 48 \text{ ppm} \ll u_r = 7600 \text{ ppm}$, the formula is well within current experimental bounds.

Proton-to-electron mass ratio.

$$\frac{m_p}{m_e} = 6\pi^5 = 6 \times 305.020\dots = 1836.118\dots \quad (7)$$

The CODATA 2022 value is $\mu = 1836.152\,673\,426(32)$ ($u_r = 1.7 \times 10^{-11}$). The deviation is $\delta = -19$ ppm. Since $u_r = 17$ ppt, the formula deviates by $|\delta|/u_r \approx 10^6$ times the experimental uncertainty. This formula is therefore **already falsified as exact** by current measurements, but it remains a striking approximation accurate to 5 significant figures.

3.4 Error distribution analysis

The 16 approximate formulas (excluding the two exact results and the Pellis formula) exhibit a characteristic error distribution:

- **4 formulas** with $|\delta| < 50$ ppm: m_p/m_e , α_s , $\sin^2\theta_{23}$, δ_{CP} .
- **6 formulas** with $50 \leq |\delta| < 100$ ppm: M_Z , $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, T_{CMB} , $\sin^2\theta_W$.
- **4 formulas** with $100 \leq |\delta| < 600$ ppm: M_W , M_H , V_{us} .
- **1 formula** with $|\delta| \approx 900$ ppm: G .

The median deviation is approximately 85 ppm. Notably, the electroweak boson masses (M_Z , M_W) and the Higgs mass (M_H) all appear in the 100–200 ppm range, suggesting that the monomial framework captures the leading-order structure of electroweak symmetry breaking but misses higher-order corrections.

3.5 Precision versus breadth

The two frameworks occupy complementary positions in the precision–breadth trade-off:

- **Pellis (polynomial):** Achieves $\delta = -0.09$ ppb for a single constant (α^{-1}) using three terms. The polynomial structure allows each successive term to “correct” the previous approximation, analogous to a perturbative series.
- **Trinity (monomial):** Achieves $|\delta| < 100$ ppm for ~ 10 constants simultaneously using a single-term template. The monomial constraint sacrifices per-formula precision for a unified algebraic structure.

This trade-off is not inherently an advantage of either approach; it reflects different design philosophies. Polynomial frameworks are naturally suited to high-precision approximation of individual constants, while monomial frameworks are suited to identifying common mathematical structure across many constants.

4 Discussion

4.1 Structural complementarity: polynomial interference vs. monomial scaling

The deepest structural observation of this comparison is the complementarity between the Pellis and Trinity approaches. In the Pellis framework, α^{-1} emerges as a sum of three terms that partially cancel (interference): $137.508 - 0.472 + 0.000291 = 137.036\dots$. In the Trinity framework, the strong coupling emerges as a pure product (scaling): $4 \times 2.618 / (9 \times 9.870) = 0.1179\dots$

This mirrors a well-known pattern in physics: Fourier series (polynomial, interference) vs. power laws (monomial, scaling) often describe the same phenomenon at different levels of approximation. It is natural to ask whether the Trinity monomials are “coarse-grained” limits of underlying polynomial structures—whether each monomial formula is the leading term of a more detailed Pellis-type expansion.

4.2 Renormalization hypothesis

A speculative but testable hypothesis is that the monomial formulas correspond to renormalization-group fixed points, while the polynomial corrections capture the running between scales. In the renormalization group, coupling constants at a fixed point often take simple, scale-invariant forms (monomials in fundamental parameters), while their values at finite energy involve logarithmic corrections (which could manifest as additive polynomial terms in φ^{-n}).

This hypothesis would predict that: (a) constants known at high precision (α^{-1} , μ) should admit polynomial improvements over their monomial approximations, and (b) constants known only at low precision (G , α_s) should currently be indistinguishable from their monomial forms. The existing data are consistent with this prediction but do not constitute evidence for it.

4.3 Can π and e emerge from φ -recursive processes?

A deeper question is whether the mathematical constants π and e appearing in the Sacred Formula are themselves derivable from φ -recursive processes. Several mathematical identities connect these constants:

$$\pi = 4 \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}, \quad e = \sum_{k=0}^{\infty} \frac{1}{k!}, \quad (8)$$

and both arise naturally in the theory of continued fractions, where φ plays a distinguished role as the “most irrational” number (slowest convergence of continued fraction approximants). If a single φ -recursive dynamical system could generate π , e , and γ as natural outputs, the Sacred Formula would reduce to a structure with a single free parameter (φ).

4.4 The G problem

The gravitational constant G presents a unique challenge. Its CODATA 2022 relative uncertainty is $u_r = 2.2 \times 10^{-5}$ (22,000 ppm)—three to six orders of magnitude larger than other constants in our table. The Trinity formula $G = \pi^3 \gamma^2 / \varphi$ deviates by ~ 900 ppm, which is well within experimental uncertainty. However, this means that *any* formula producing a value within 2% of the measured G would be equally “consistent” with experiment. The G formula cannot be meaningfully tested until gravitational measurements improve by at least two orders of magnitude.

4.5 What would constitute falsification

An empirical formula V_{formula} is falsified when either:

- (a) A new CODATA adjustment shifts the central value such that the formula’s deviation increases beyond a significance threshold, or
- (b) Improved measurements reduce the experimental uncertainty below $|\delta|$, proving that the formula’s prediction is inconsistent with experiment.

Several formulas in our table are already in tension with experiment by criterion (b):

- $m_p/m_e = 6\pi^5$ has $|\delta| = 19$ ppm but $u_r = 17$ ppt—the formula is excluded as exact at the $10^6\sigma$ level.

- $M_Z = 7\pi^4\varphi^3/243$ has $|\delta| = 61$ ppm but $u_r = 23$ ppm—the formula deviates at the $\sim 2.7\sigma$ level.

Conversely, formulas such as $\alpha_s = 4\varphi^2/(9\pi^2)$ ($|\delta| = 48$ ppm, $u_r = 7600$ ppm) and $G = \pi^3\gamma^2/\varphi$ ($|\delta| \sim 900$ ppm, $u_r = 22,000$ ppm) remain well within experimental bounds and cannot currently be distinguished from coincidence.

The Pellis formula $\alpha^{-1} = 360\varphi^{-2} - 2\varphi^{-3} + (3\varphi)^{-5}$ is the most stringently tested: its $|\delta| = 0.089$ ppb is within $u_r = 0.16$ ppb, but future improvements in the electron anomalous magnetic moment measurement (which determines α) may either confirm or exclude it at the ppb level.

Epistemological note

Disclaimer: The formulas presented in this paper are empirical approximations. They reproduce CODATA 2022 values to varying degrees of precision but do not constitute derivations from first principles. No Lagrangian, renormalization-group equation, or symmetry argument from which the Sacred Formula $V = n \cdot 3^k \pi^m \varphi^p e^q \gamma^r$ follows has been identified. Close numerical agreement does not imply causal or structural necessity. The standard of proof in physics requires a predictive framework that (a) derives the formula from a deeper physical principle, (b) predicts new observables not used in fitting, and (c) withstands improved experimental precision. This paper makes no such claim and invites critical evaluation.

5 Conclusion

We have systematically compared 18 empirical formulas from the Trinity monomial framework and the Pellis polynomial framework against CODATA 2022 recommended values for fundamental physical constants. The Trinity identity $\varphi^2 + \varphi^{-2} = 3$ is algebraically exact. Sixteen additional formulas achieve deviations below 1000 ppm; ten fall below 100 ppm. The Pellis three-term formula for α^{-1} achieves $\delta = -0.09$ ppb, within the current experimental uncertainty of 0.16 ppb.

We emphasize that these results constitute **empirical approximations, not physical derivations**: no Lagrangian or symmetry principle from which the Sacred Formula follows has been identified. The structural complementarity between polynomial and monomial representations—additive interference versus multiplicative scaling—merits further mathematical investigation. Whether these patterns reflect deep structure in the fundamental constants or are elaborate numerical coincidences remains an open question that only improved experimental precision and theoretical insight can resolve.

Data availability. All Python verification scripts are available at <https://github.com/trinity-project>.

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A Full Formula Catalog

Table 4 lists all 18 formulas analyzed in this work with their symbolic forms, exponent decompositions, predicted values, CODATA 2022 values, and relative deviations.

Table 4: Complete catalog of formulas analyzed in this work. Tier 0 formulas are algebraically exact. For the Pellis formula (P1), the polynomial structure does not conform to the monomial template.

#	Tier	Formula	Predicted	CODATA 2022	δ
1	0	$N_{\text{gen}} = \varphi^2 + \varphi^{-2}$	3	3 gen.	EXACT
2	0	$\theta_{\text{QCD}} = \varphi^2 + \varphi^{-2} - 3 $	0	$< 10^{-10}$	EXACT
3	1	$\delta_{\text{CP}} = 8\pi^3/(9e^2)$	3.72999 rad	3.73 rad	+1.6 ppm
4	1	$m_p/m_e = 6\pi^5$	1836.118	1836.15267(3)	-19 ppm
5	1	$\sin^2\theta_{23}$	0.5460	0.546	+28 ppm
6	1	$\alpha_s = 4\varphi^2/(9\pi^2)$	0.11789	0.1179(9)	-48 ppm
7	1	$M_Z = 7\pi^4\varphi e^3/243$	91.193 GeV	91.1876(21)	+61 ppm
8	1	$\sin^2\theta_{12}$	0.30702	0.307	+75 ppm
9	1	$\sin^2\theta_{13}$	0.02200	0.0220	+76 ppm
10	1	$\sin^2\theta_W = 2\pi^3 e/729$	0.23123	0.23121(4)	+92 ppm
11	1	$T_{\text{CMB}} = \frac{5\pi^4\varphi^5}{729e}$	2.72575 K	2.72548(57)	+98 ppm
12	2	$M_W = 162\varphi^3/(\pi e)$	80.359 GeV	80.369(13)	-127 ppm
13	2	$M_H = 135\varphi^4/e^2$	125.23 GeV	125.20(11)	+210 ppm
14	2	$V_{us} = 3\gamma/\pi$	0.22543	0.22530	+570 ppm
15	2	$G = \pi^3\gamma^2/\varphi$	~ 6.674	6.67430(15)	~ 900 ppm
P1	—	$\alpha^{-1} = 360\varphi^{-2} - 2\varphi^{-3} + (3\varphi)^{-5}$	137.035999165	137.035999177(21)	-0.09 ppb

B Computational Details

The following Python script was used to verify all numerical values in this paper. It uses the decimal module with 50-digit precision to avoid floating-point rounding errors.

```

from decimal import Decimal, getcontext
getcontext().prec = 50

sqrt5 = Decimal(5).sqrt()
phi = (1 + sqrt5) / 2
pi = Decimal('3.14159265358979323846264338327950288419716939937510')
e = Decimal('2.71828182845904523536028747135266249775724709369995')
gamma = phi**(-3)

CODATA = {
    'alpha_inv': Decimal('137.035999177'),
    'mu': Decimal('1836.152673426'),
    'alpha_s': Decimal('0.1179'),
    'sin2_thetaW': Decimal('0.23121'),
    'T_CMB': Decimal('2.72548'),
    'M_H': Decimal('125.20'),
    'M_Z': Decimal('91.1876'),
    'M_W': Decimal('80.369'),
}

def delta_ppm(formula_val, codata_val):
    return float((formula_val - codata_val) / codata_val) * 1e6

# Trinity identity
print(f"phi^2 + phi^-2 = {phi**2 + phi**(-2)}") # 3.000...

# Pellis alpha^-1

```

```

pellis = 360*phi**(-2) - 2*phi**(-3) + (3*phi)**(-5)
print(f"Pellis alpha^-1 = {pellis}")
print(f"delta = {delta_ppm(pellis, CODATA['alpha_inv'])*1e3:.3f} ppb")

# Trinity formulas
formulas = {
    'mp/me = 6*pi^5': (6*pi**5, 'mu'),
    'alpha_s = 4*phi^2/(9*pi^2)': (4*phi**2/(9*pi**2), 'alpha_s'),
    'sin2_thetaW': (2*pi**3*e/729, 'sin2_thetaW'),
    'T_CMB': (5*pi**4*phi**5/(729*e), 'T_CMB'),
    'M_Z': (7*pi**4*phi*e**3/243, 'M_Z'),
    'M_H': (135*phi**4/e**2, 'M_H'),
    'M_W': (162*phi**3/(pi*e), 'M_W'),
}
for name, (val, key) in formulas.items():
    d = delta_ppm(val, CODATA[key])
    print(f"{name}: {float(val):.10f}, delta = {d:.1f} ppm")

```

C CODATA 2022 Source Data

Table 5 lists all CODATA 2022 values used in this work. All values are from the NIST CODATA 2022 database [2].

Table 5: CODATA 2022 recommended values used in this work. Source: NIST [2], <https://physics.nist.gov/cuu/Constants/>.

Constant	Symbol	CODATA 2022 Value	Rel. Uncertainty
Fine-structure constant (inv.)	α^{-1}	137.035 999 177(21)	1.6×10^{-10}
Proton/electron mass ratio	$\mu = m_p/m_e$	1836.152 673 426(32)	1.7×10^{-11}
Gravitational constant	G	$6.67430(15) \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$	2.2×10^{-5}
Strong coupling	$\alpha_s(M_Z)$	0.1179(9)	7.6×10^{-3}
Weak mixing angle	$\sin^2 \theta_W$	0.23121(4)	1.7×10^{-4}
CMB temperature	T_{CMB}	2.72548(57) K	2.1×10^{-4}
Higgs boson mass	M_H	125.20(11) GeV	8.8×10^{-4}
Z boson mass	M_Z	91.1876(21) GeV	2.3×10^{-5}
W boson mass	M_W	80.369(13) GeV	1.6×10^{-4}

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